

UTSAVKUMAR VAGHANI

Senior Backend Engineer | Distributed Systems | Payments and Production GenAI

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Summary

Backend engineer, 5+ years in fintech (Go, Node.js, Python): payments infrastructure across four gateways, investment alerting for 5M users, and the order-management core of a multi-market wealth platform. Shipped two production GenAI systems at Razorpay (70% containment; INR 50L/month cost removed) while leading a team of five.

Highlights

- **Payments PaaS from scratch:** 3 payment types across Razorpay, Juspay, Apple IAP and Zerodha; USD 1.5M+/yr
- **Investment alerting for 5M users (1M active):** pre-market, intraday and end-of-day alerts, built from scratch
- **Production GenAI at Razorpay:** 300–400 auto-resolved tickets/day, INR 50L/month support cost removed; led a team of 5

Experience

Keenai Global

Aug 2025 – Present

Senior Associate, Product Engineering (Backend Engineer)

Bangalore, India

- Built the **order-management service** as the foundation for **all instrument types** — equities, ETFs, mutual funds, options, FCNs, hedge funds — across **US, Singapore and other exchanges**, with **Temporal-orchestrated** workflows; **cut list-orders latency by ~10s**.
- Built the custodian **execution-report pipeline** on SQS pub/sub (**per-order FIFO**, **dedupe** on execution IDs, batching, **Redlock** locks) and the **org-wide domain-events service** with **Kafka** (**MSK**, **IAM**), **FIFO** partition keys and **DLQs**.
- Enabled **real-time portfolio data**: OMS → portfolio-service sync on order fulfilment, **chained jobs with Postgres debounce** and **cross-account EventBridge**; reconciled against bank holdings.
- Led the **portfolio revamp** replacing a **third-party analytics vendor with in-house services** — overview, holdings, allocation, performance (TWRR, XIRR, AUM) — from spec to go-live; **own the portfolio module** for the wealth platform.
- Designed a **price-hierarchy engine** selecting the valuation price per instrument across market-data sources; **cut reconciliation breaks by 20%**.
- Built **portfolio versioning (business-day + intraday)** over **100K+ transactions per family office**: SFO/entity-level **Ops dashboard** (versioned-vs-live views, **audit logging**), **auto-versioning on successful reconciliation**.
- Built the **reporting module end-to-end**: async **Handlebars-templated** report-generation pipeline for transaction and holdings reports, and a **drag-and-drop template builder** for RM, IC and Pulse-client use cases with **per-tenant branding**.
- Stack: Node.js (TypeScript), Java, PostgreSQL, DynamoDB, Redis, Kafka (MSK), SQS, EventBridge, Temporal, AWS Lambda

Razorpay

Sep 2024 – Aug 2025

Senior Software Engineer

Bangalore, India

- Built **Yogi, a GenAI system** that auto-resolves merchant support tickets: **300–400 resolutions/day** and **INR 50L/month reduction in human-agent cost**; drove CSAT for automated resolutions.
- Built a **RAG + LLM merchant-onboarding assistant** (intent classification, SOP-driven flows): **70% containment**; instrumented **containment, accuracy, latency and cost-per-ticket metrics** for safe rollout.
- **Led and mentored 5 engineers**; owned availability, on-call and runbooks; **unblocked release cadence** via a dedicated deployment of the legacy central service (weekly → on-demand deploys).
- Stack: Python (FastAPI, Celery), OpenAI APIs, RAG, MCP, n8n

Smallcase

Mar 2022 – Sep 2024

SDE I → SDE II

Bangalore, India

- Built a **payments Platform-as-a-Service from scratch** — one-time, subscription and token-based recurring payments across **Razorpay, Juspay, Apple In-App Purchase and Zerodha, USD 1.5M+ per year** — powering digital-gold and membership recurring payments with idempotent retries and reconciliation (Go, NestJS, MongoDB, AWS).
- Built **investment alerting from scratch** for **5M users (1M active)**: pre-market, intraday and end-of-day alerts.
- Made legacy **Zerodha (Kite) transaction flows fault-tolerant** (idempotent retries, reconciliation) and **revamped the offer engine** for digital gold and membership purchases.

Zopsmart

Jan 2021 – Feb 2022

SDE Intern → SDE I

Bangalore, India

- Built a **Go application framework** (web and CLI) with generic interfaces for pub/sub (Amazon SNS), metrics (Prometheus) and AWS services; **adopted org-wide**.
- Built **CSP** (credential service provider) for authenticated inter-service communication; end-to-end APIs for a marketplace application.

Skills

- **Languages:** Go (Golang), JavaScript/TypeScript (Node.js), Python
- **Backend:** gRPC, REST, NestJS, Gofiber, FastAPI, Celery, Kafka, SNS/SQS, Temporal, event-driven
- **GenAI:** OpenAI APIs, RAG, MCP, n8n, LLM evaluation and observability
- **Data:** PostgreSQL, MongoDB, DynamoDB, MySQL, Redis
- **Cloud and infra:** AWS (EC2, S3, Lambda, SNS, SQS), Docker, Prometheus, CI/CD, distributed systems

Education and Certifications

- **BE, Computer Engineering** — G H Patel College of Engineering & Technology, 2017–21, CGPA 8.22 · **AWS Certified Cloud Practitioner**